

Useful Memory Has a Horizon

Structural Worst-Case Limits for Finite-Memory Tracking under Drift

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Abstract

Under drift, memory is both a statistical resource and a temporal liability: retaining more past information reduces variance, but it also increases the age of the evidence used to represent the present. The central question is therefore not just how much to remember, but how long remembered evidence should remain operationally valid.

We study this trade-off in a worst-case Wasserstein envelope. For the finite-memory averaging class analyzed here, the tracking error decomposes into a decreasing finite-sample term and an increasing staleness term. In the linear staleness regime induced by the local W_2 -Lipschitz path bound, this yields a **cube-root horizon law**: the optimal memory scale obeys $n^* \asymp (C_K/\zeta)^{2/3}$, and the minimum tracking error obeys $\mathcal{E}_{\min} \asymp C_K^{2/3} \zeta^{1/3}$. In the critical-window regime, we also prove a minimax lower bound, showing that this finite-memory floor is structural rather than estimator-specific.

Empirically, Gaussian drift experiments recover the predicted U -curve and a finite useful-memory optimum. Continuous-drift benchmarks expose a gap between temporal validity and changepoint evidence; alternating-timescales benchmarks show that contraction is easier than re-expansion; and horizon-misalignment experiments show that stale memory can carry a nonlinear predictive cost.

The resulting picture is operational: temporal validity is not changepoint evidence, and conservative horizon control can prevent stale memory from persisting before a detector has enough evidence to alarm.

Keywords: tracking under drift; useful memory; temporal validity; Wasserstein distance; Sinkhorn divergence; concept drift; adaptive memory.

MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore answer a basic but easy-to-miss question: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. More past data reduces sampling noise, but it also increases the age of the evidence used to describe the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The resulting problem is a worst-case horizon problem, not a detector comparison.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory and a staleness term that increases with drift. In the linear staleness regime induced by the local W_2 -Lipschitz path bound, optimizing that balance yields a cube-root law: $n^* \asymp (C_K/\zeta)^{2/3}$ and $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$. The same class also admits a minimax lower bound in the critical-window regime, so the finite-memory floor is structural rather than a quirk of one estimator.

The central conceptual distinction is simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained memory is already stale for the present. That separation creates the operational gap that makes a conservative horizon cap useful even when no detector has fired.

The contributions are fivefold:

- It formalizes useful memory under drift as a variance-staleness trade-off in Wasserstein geometry.
- It proves a worst-case finite-memory horizon law under local W_2 -Lipschitz drift.
- It shows a Gaussian-location lower bound establishing a structural tracking floor in the critical-window regime.
- It distinguishes temporal validity from changepoint evidence and identifies a cap-only regime.
- It instantiates the theory as a conservative horizon regulator and evaluates where it helps, saturates, and fails.

The empirical evidence follows that logic in stages. It first recovers the predicted U -curve and the finite useful-memory optimum. It then shows the gap between temporal validity and changepoint evidence, the cap-only regime, the recovery asymmetry under alternating drift, and the nonlinear cost of horizon misalignment. ELEC2 shows the same qualitative issue outside the synthetic setting, while additional stability and sensitivity checks appear in the appendix.

The theoretical results are stated in W_2 ; the Sinkhorn divergence is the computational measurement layer used in the synthetic experiments. The window size result of [Hinder et al. \[2023\]](#) motivates the broader issue, but the contribution here is more specific: memory has a finite temporal validity horizon under worst-case drift.

2 Finite Memory Under Drift

Under continuous drift, memory is both a statistical resource and a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

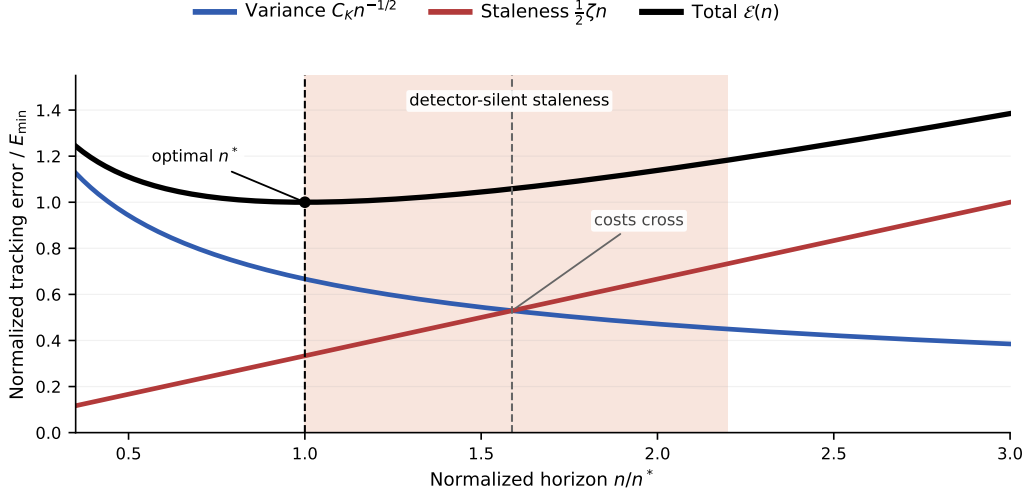


Figure 1: Two clocks of drift. The vertical axis is tracking error and the horizontal axis is the memory horizon, shown in normalized theorem units. More memory reduces the finite-sample variance term but increases staleness, producing a U-shaped total error. The optimum n^* occurs before the point where the two cost levels cross, because the optimum is determined by marginal balance rather than equality of the two cost values. The shaded region marks detector-silent staleness.

2.1 Variance-Staleness Decomposition

Theorem 2.1 (Lag-Inclusive Tracking Error Bound). *Let $X_{t-j} \sim P_{t-j}^*$ and let*

$$\hat{P}_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{X_{t-j}}$$

be the uniform-window empirical estimator. Assume the target path is locally W_2 -Lipschitz with rate ζ , so that

$$W_2(P_{t-j}^*, P_t^*) \leq \zeta j \quad \text{for all } 0 \leq j < n,$$

and assume the finite-sample term obeys

$$\mathbb{E} \left[W_2 \left(\hat{P}_t^{(n)}, \bar{P}_t^{(n)} \right) \right] \leq C_K n^{-1/2},$$

where $\bar{P}_t^{(n)} := n^{-1} \sum_{j=0}^{n-1} P_{t-j}^$. Then the expected tracking error satisfies*

$$\mathcal{E}(n) \leq C_K n^{-1/2} + \frac{1}{2} \zeta n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

Lemma 2.2 (Finite-sample Wasserstein term). *Let Q be a probability measure on $\mathbb{R}^{d_{\text{eff}}}$ with support contained in a ball of radius R and finite second moment. Let $X_1, \dots, X_n$ be i.i.d. samples from Q , and let $\hat{Q}_n := n^{-1} \sum_{i=1}^n \delta_{X_i}$. Then:*

- If $d_{\text{eff}} < 4$, there exists a constant $C_{d_{\text{eff}}, R}$ such that

$$\mathbb{E} W_2(\hat{Q}_n, Q) \leq C_{d_{\text{eff}}, R} n^{-1/2}.$$

- If $d_{\text{eff}} = 4$, the same bound holds with an additional $(\log n)^{1/2}$ factor.
- If $d_{\text{eff}} > 4$, the raw Wasserstein rate is slower than $n^{-1/2}$, so the constant-only form above should not be used.

In that regime, the statistical fluctuations do not follow the $n^{-1/2}$ carrier exponent, so the resulting horizon scaling stays outside the displayed cube-root regime. If the geometry is computed with debiased Sinkhorn divergence S_ε on a bounded domain of diameter D , then for fixed $\varepsilon > 0$ the empirical fluctuations remain $O(n^{-1/2})$ with a constant $L(D, \varepsilon)$ that is dimension-free at fixed ε and increases as $\varepsilon \downarrow 0$.

A short numerical sanity check for this regime appears in Appendix D; it is not used as a theorem, but it matches the intended low-dimensional interpretation of the finite-sample term.

2.2 A Structural Worst-Case Horizon

Remark 2.3 (Scope of the horizon law). The cube-root law follows when the finite-sample term behaves as $\mathbb{E} W_2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-1/2}$. The discussion below uses only that regime. More generally, if the finite-sample term admits the scaling

$$\mathbb{E} W_2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-a} \quad \text{for some } a > 0,$$

then balancing it against the drift-induced staleness cost $\frac{1}{2}\zeta n$ yields

$$n^*(a) = \left(\frac{2a C_K}{\zeta} \right)^{\frac{1}{a+1}}, \quad \mathcal{E}_{\min}(a) = \frac{a+1}{(2a)^{a/(a+1)}} C_K^{1/(a+1)} \zeta^{a/(a+1)}.$$

In the setting of lemma 2.2, the effective exponent is $a = 1/2$ for $d_{\text{eff}} < 4$, with an additional $(\log n)^{1/2}$ factor for $d_{\text{eff}} = 4$; for $d_{\text{eff}} > 4$ the raw Wasserstein rate is slower than $n^{-1/2}$. That case is outside the displayed cube-root regime.

The two terms in theorem 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic variance-staleness trade-off that governs useful memory under drift.

Proposition 2.4 (Optimal Window Under Drift). *Under theorem 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* = \left(\frac{C_K}{\zeta} \right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}.$$

Proof. Apply the triangle inequality for W_2 :

$$W_2\left(\widehat{P}_t^{(n)}, P_t^*\right) \leq W_2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + W_2\left(\bar{P}_t^{(n)}, P_t^*\right).$$

The first term is controlled by the assumed finite-sample bound. For the second term, let π_j be an optimal coupling between P_{t-j}^* and P_t^* . Then the averaged coupling $\bar{\pi} := n^{-1} \sum_{j=0}^{n-1} \pi_j$ couples $\bar{P}_t^{(n)}$ with P_t^* , so the transport cost is at most the average of the individual costs. Combined with the W_2 -Lipschitz drift bound, this gives

$$W_2\left(\bar{P}_t^{(n)}, P_t^*\right) \leq \frac{1}{n} \sum_{j=0}^{n-1} W_2(P_{t-j}^*, P_t^*) \leq \frac{1}{n} \sum_{j=0}^{n-1} \zeta j \leq \frac{1}{2} \zeta n.$$

Combining the two bounds yields the stated inequality. $\square$

Remark 2.5 (Measurement layer). The theorem is stated in W_2 , so ε does not enter the mathematical drift law itself. In the synthetic measurement layer, however, the observable geometry is often a debiased Sinkhorn divergence with fixed ε , which makes the finite-sample constant calibration-dependent. In the synthetic benchmarks, $\hat{\zeta}_t$ is estimated from short-window block differences using an EMA update and C_K is calibrated from a stationary prefix; Appendix E records the sensitivity check and the prequential calibration rule.

For fixed ε on bounded domains, the debiased Sinkhorn divergence converges at rate $n^{-1/2}$ with ε -dependent constants [Genevay et al., 2019], while the approximation gap to W_2^2 is $O(\varepsilon)$ and vanishes as $\varepsilon \rightarrow 0$. We therefore treat Sinkhorn as a calibration proxy rather than part of the law itself.

Corollary 2.6 (EWMA Analogue). *For exponential weights $w_j = \alpha(1 - \alpha)^j$, the effective lag is $\tau_\alpha = (1 - \alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

2.3 Temporal Validity and Memory Age

Definition 2.7 (Effective Memory Age). Let an estimator place weights $w_0, \dots, w_{n-1}$ on the most recent observations, with $\sum_j w_j = 1$. Its *effective memory age* is

$$A(w) = \sum_{j=0}^{n-1} j w_j,$$

and its corresponding freshness proxy is the monotone transform $F(w) = 1/(1 + A(w))$. The role of $F(w)$ is purely geometric: it records how old the retained evidence is before any local drift scale is specified. For a uniform window, $A(w) = (n - 1)/2$.

Definition 2.8 (Temporal Validity). For an estimator with weights w at time t , define its *operational staleness*

$$S_t(w) = \sum_{j=0}^{n-1} w_j W_2(\mathcal{P}^*(S, t - j), \mathcal{P}^*(S, t)).$$

Its *temporal validity* is the inverse alignment score

$$C_t(w) = \frac{1}{1 + S_t(w)}.$$

If the target path is locally W_2 -Lipschitz at rate ζ_t , in the sense that $W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t)) \leq \zeta_t j$ for the relevant lags, then

$$S_t(w) \leq \zeta_t A(w), \quad C_t(w) \geq \frac{1}{1 + \zeta_t A(w)}.$$

In that sense, $F(w)$ is the rate-free freshness proxy, while $C_t(w)$ measures operational validity with the current distribution.

The quantity $A(w)$ gives a compact description of the temporal age of a finite-memory estimator. Short windows preserve temporal validity at the cost of variance; long windows reduce variance at the cost of staleness. The cube-root horizon marks the point where that exchange stops improving. For a uniform window under approximately constant local drift, the induced staleness scales as $S_t(w) \approx \zeta(n-1)/2$, which is the same age term that appears in theorem 2.1 up to constants.

2.4 A Structural Finite-Memory Floor

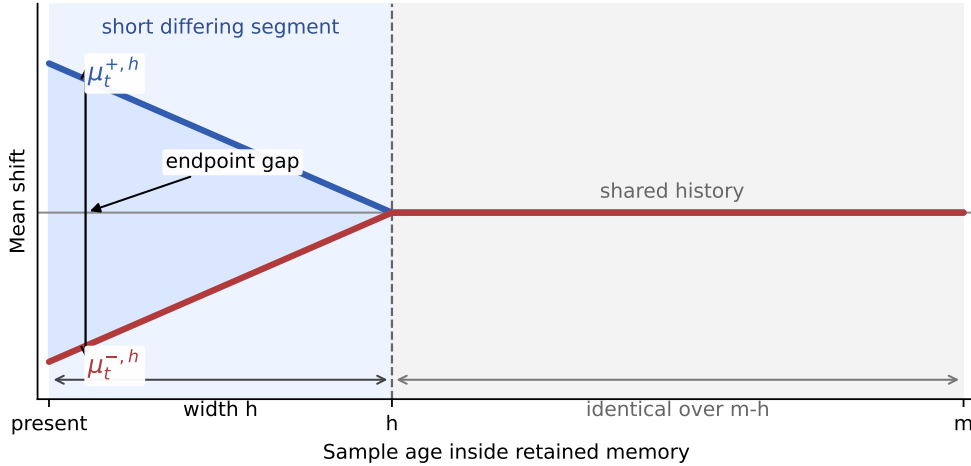


Figure 2: Lower-bound witness on the Gaussian location subclass. The vertical axis is the mean shift and the horizontal axis is sample age inside the retained memory. The two paths are far apart at the present but differ only over a short recent segment of width h , while they are identical over the remaining retained history. This is the structural obstruction behind the finite-memory floor.

Proposition 2.9 (Finite-Memory Floor via Gaussian Location Subclass). *Let $\mathcal{P}_{\zeta,m}$ be the class of length- m distribution paths for which the following hold:*

1. *the path is W_2 -Lipschitz with rate at most ζ ;*

2. the lower-bound construction is carried out on the Gaussian location subclass $X_t \sim \mathcal{N}(\mu_t, \sigma^2 I_d)$ with $\|\mu_t - \mu_s\|_2 \leq \zeta|t - s|$;
3. the estimator at time m only uses the last m samples.

Measure risk by the endpoint W_2 error, which coincides with Euclidean endpoint error in the Gaussian location subclass. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E} d(\hat{\mathcal{P}}_m, \mathcal{P}^*) \gtrsim \max\{C_K m^{-1/2}, c \sigma^{2/3} \zeta^{1/3}\}.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$. We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in the admissible drift class. For the two hypotheses $\theta_{\pm} = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\text{KL}(P_+, P_-) = \frac{1}{2\sigma^2} \sum_{j=1}^m (\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}} \right)^2 = \frac{1}{2}.$$

For any estimator δ_m , the plug-in test $\psi = \mathbf{1}\{\delta_m \geq 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam’s two-point lemma and Pinsker’s inequality, the average testing error is at least

$$\frac{1 - \text{TV}(P_+, P_-)}{2} \geq \frac{1 - \sqrt{\text{KL}(P_+, P_-)/2}}{2} \geq \frac{1}{4}.$$

Hence

$$R_m^* \geq \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \leq h \leq m$ and define

$$\beta_h := \min\left\{\zeta, \frac{\sigma}{4h^{3/2}}\right\}, \quad \mu_{t-j}^{\pm, h} := \pm \beta_h (h - j)_+, \quad 0 \leq j \leq m - 1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to the admissible class because its slope magnitude is at most $\beta_h \leq \zeta$. The endpoint gap is

$$|\mu_t^{+, h} - \mu_t^{-, h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\text{KL}(P_{+, h}, P_{-, h}) = \frac{1}{2\sigma^2} \sum_{j=0}^{m-1} (2\beta_h (h - j)_+)^2 \leq \frac{2\beta_h^2}{\sigma^2} \sum_{r=1}^h r^2 \leq \frac{2\beta_h^2}{\sigma^2} h^3 \leq \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\beta_h h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1 - \text{TV}(P_{+,h}, P_{-,h})}{2} \geq \frac{1 - \sqrt{\text{KL}(P_{+,h}, P_{-,h})/2}}{2} \geq \frac{3}{8}.$$

Therefore

$$R_m^* \geq \frac{3}{8} \beta_h h \geq \frac{3}{32} \min\{\zeta h, \sigma h^{-1/2}\}$$

which is again a fully explicit numerical lower bound. Since h was arbitrary,

$$R_m^* \geq \frac{3}{32} \sup_{1 \leq h \leq m} \min\{\zeta h, \sigma h^{-1/2}\}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \leq h_* := (\sigma/\zeta)^{2/3} \leq m$. Choose an integer $\bar{h} \in [h_*/2, h_*]$. Then

$$\zeta \bar{h} \geq \frac{1}{2} \zeta h_* = \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and

$$\sigma \bar{h}^{-1/2} \geq \sigma h_*^{-1/2} = \sigma^{2/3} \zeta^{1/3}.$$

Therefore

$$\min\{\sigma \bar{h}^{-1/2}, \zeta \bar{h}\} \geq \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \geq \frac{3}{64} \sigma^{2/3} \zeta^{1/3}.$$

So one admissible explicit constant in the critical-window regime is $c = 3/64$.

Remark 2.10 (Scope of the lower bound). The lower bound is parametric only in the testing construction: it uses a Gaussian location subclass whose mean path is itself W_2 -Lipschitz, and the distance in the lower bound is the endpoint W_2 error (Euclidean on that subclass). That is enough to make the result minimax for any larger class containing that subclass, including the ambient W_2 -Lipschitz path class. So the proposition is not claiming that every path is Gaussian; it is claiming that the worst case already appears inside the Gaussian mean-shift subproblem. Equivalently, since the Gaussian location family is a subfamily of the ambient W_2 -Lipschitz class at the same drift rate ζ , any lower bound proven on that subclass is automatically a lower bound for the larger class.

Remark 2.11 (Measurement layer). Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Evidence

The experiments test a single claim: under continuous drift, memory must be regulated by temporal validity rather than by statistical evidence of abrupt change

alone. The evidence separates three layers that are often collapsed: the finite-memory horizon itself, the conflict between temporal validity and detector evidence, and the predictive cost of horizon misalignment.

In the experiments, UMR is a backend-agnostic conservative horizon cap driven by a smoothed local drift proxy. It is evaluated in two roles. First, it acts as a runtime horizon policy for estimators whose effective memory can be adjusted, such as capped sliding-window, EWMA, and multi-horizon backends. Second, it is wrapped around ADWIN as one detector-side instance of the same temporal-validity rule. Fixed-window methods remain uncapped baselines. The resulting comparisons make detector-silent staleness visible.

3.1 Structural U-Curve

We begin with the Gaussian sweep used to validate the finite-memory trade-off. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum remains finite and moves in the expected direction as drift strengthens, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

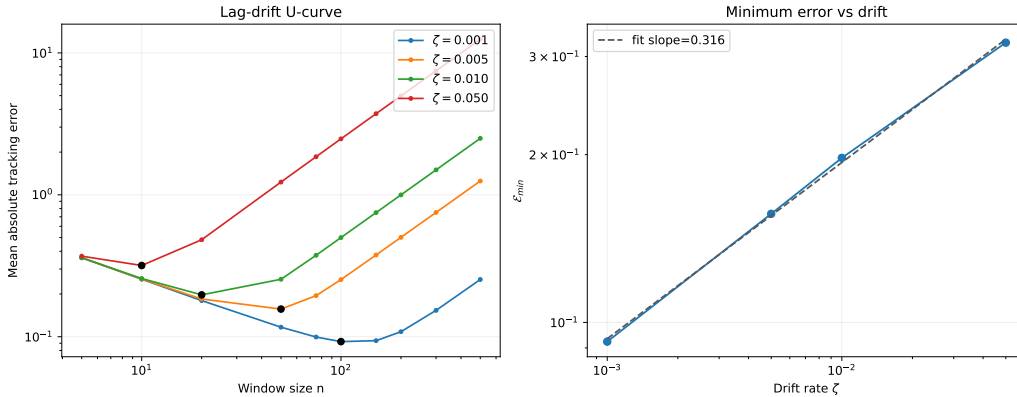


Figure 3: Finite-memory law validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error decreases as drift weakens, consistent with the variance-staleness trade-off. The figure shows that the useful-memory optimum is finite and structural rather than a tuning artifact.

The same pattern is visible from the lag-variance frontier in fig. 4, which exposes the frontier knee where extra memory stops paying off. Bootstrap intervals in Appendix C support the same ordering.

3.2 Temporal Validity Is Not Changepoint Evidence

The central empirical contrast is the gap between two questions. The detector asks whether there is enough statistical evidence of a change. The horizon asks whether the retained evidence is still temporally valid for the present. Those questions need not be synchronized.

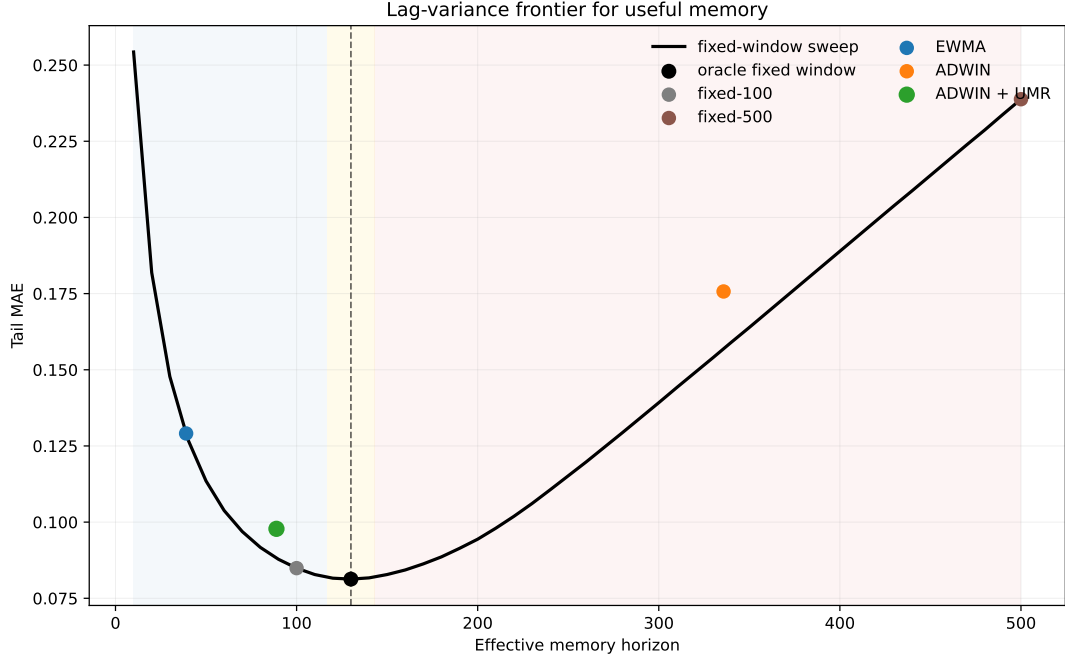


Figure 4: Lag-variance frontier for useful memory. The fixed-window sweep traces the empirical frontier: short horizons are noisy, long horizons are stale, and the knee marks the useful-memory optimum. The frontier is asymmetric because the gain from more memory saturates while the cost of age keeps accumulating.

The gap becomes concrete on the continuous-drift benchmark. We compare a short fixed window, a long fixed window, EWMA, ADWIN, and an ADWIN wrapper equipped with the UMR horizon cap. ADWIN can retain far more memory than the useful-memory scale, while a conservative cap contracts that stale window earlier.

Table 1: Continuous-drift benchmark. The key gap is between statistical evidence and temporal validity: ADWIN keeps a much wider window than the UMR-capped horizon scale, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Detector events	Cap events
Fixed-100	0.0849	100.0	—	—
Fixed-500	0.2388	500.0	—	—
EWMA	0.1291	39.0	—	—
ADWIN	0.1757	335.8	3.9	0.0
ADWIN + UMR	0.0978	88.9	0.0	1374.7

3.3 Recovery Asymmetry Under Alternating Drift

The next benchmark makes local horizon instability visible. The stream alternates between slow drift, fast bursts, recovery plateaus, and micro-bursts with irregular durations, so the local oracle horizon need not be globally stable. In that setting,

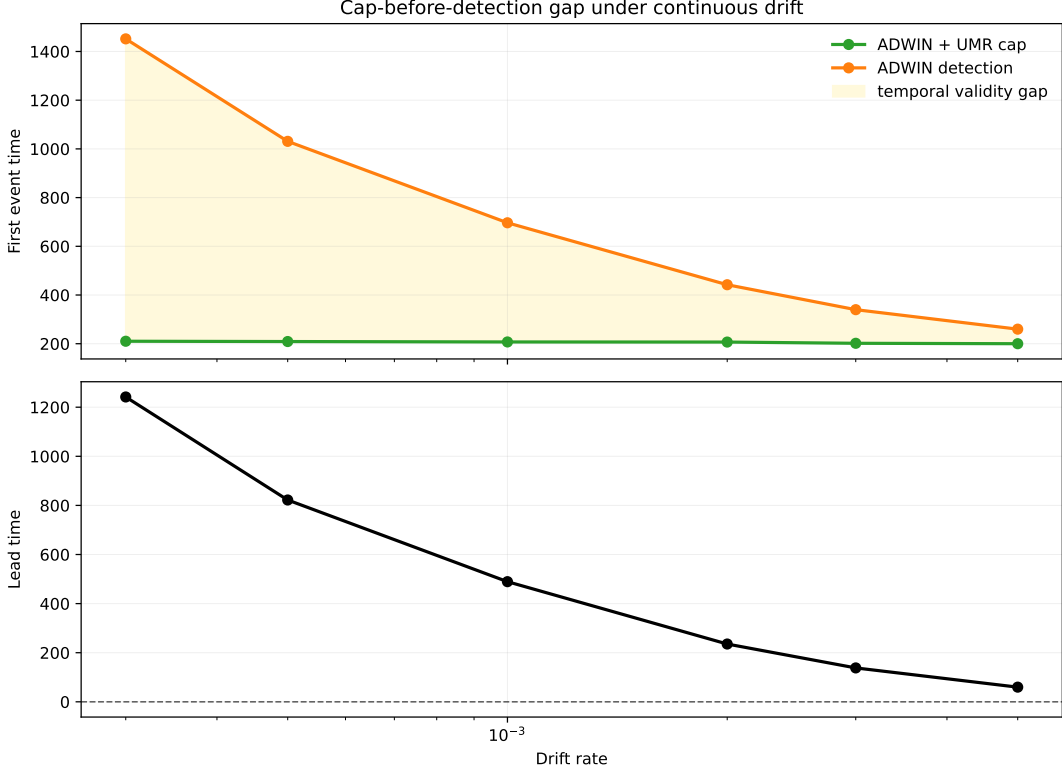


Figure 5: Temporal validity gap across drift rates. The top panel compares the first ADWIN+ UMR cap activation with the first ADWIN detection. The bottom panel shows the resulting lead time. Under slow drift, memory becomes operationally stale long before changepoint evidence is available; under faster drift, the gap narrows but remains positive.

a single fixed window can look reasonable on average while remaining locally misaligned for long stretches.

The benchmark makes two things visible. First, horizon validity can vary even when aggregate prediction error stays competitive. Second, the adjustment is asymmetric: contraction of the useful horizon is easier to trigger than re-expansion after a low-drift stretch. Proposition 3.1 formalizes the same asymmetry in the idealized EMA-to-horizon map. The transition-window table and the EMA-sensitivity sweep appear in Appendix C.

Proposition 3.1 (Asymmetric Horizon Recovery). *Let the drift estimator satisfy the EMA recursion*

$$\hat{\zeta}_t = \alpha \Delta_t + q \hat{\zeta}_{t-1}, \quad q := 1 - \alpha \in (0, 1),$$

with $\Delta_t = \delta_{\text{slow}}$ in the slow regime and $\Delta_t = M$ in the burst regime. Assume the horizon map is unclipped on $[\delta_{\text{slow}}, M]$ so that

$$n^*(\delta) = \left(\frac{C_K}{\delta} \right)^{2/3}$$

throughout the transition. Write $n_b := n^(M)$ and $n_s := n^*(\delta_{\text{slow}})$. For any $0 < \varepsilon < n_s - n_b$, define the inverse thresholds*

$$\delta_b(\varepsilon) := C_K (n_b + \varepsilon)^{-3/2}, \quad \delta_s(\varepsilon) := C_K (n_s - \varepsilon)^{-3/2}.$$

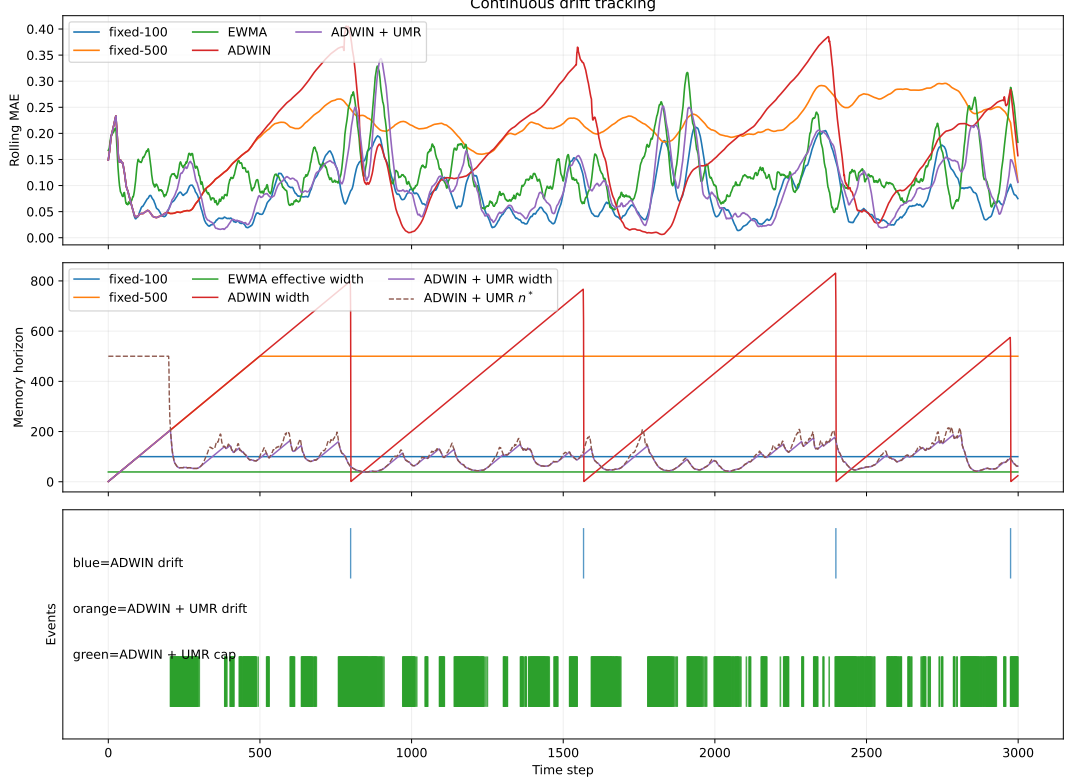


Figure 6: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; ADWIN+UMR pulls the effective horizon toward the useful-memory scale and exposes a cap-only regime in which memory becomes operationally stale before statistical changepoint evidence appears.

Then the exact recovery time from the slow regime to the burst regime is

$$T_{\text{contract}}(\varepsilon) = \left\lceil \frac{\log((M - \delta_{\text{slow}})/(M - \delta_b(\varepsilon)))}{\log(1/q)} \right\rceil,$$

while, after a burst of duration T_{burst} steps, the exact recovery time back to the slow regime is

$$T_{\text{expand}}(\varepsilon) = \left\lceil \frac{\log((\hat{\zeta}_{\text{end}} - \delta_{\text{slow}})/(\delta_s(\varepsilon) - \delta_{\text{slow}}))}{\log(1/q)} \right\rceil,$$

where

$$\hat{\zeta}_{\text{end}} = M - (M - \delta_{\text{slow}})q^{T_{\text{burst}}}$$

is the burst-end drift estimate. If

$$T_{\text{burst}} \geq T_0(\varepsilon) := \left\lceil \frac{\log(1 - \rho_\varepsilon)}{\log q} \right\rceil, \quad \rho_\varepsilon := \frac{\delta_s(\varepsilon) - \delta_{\text{slow}}}{M - \delta_b(\varepsilon)},$$

then $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$. Moreover, for small ε ,

$$M - \delta_b(\varepsilon) = \frac{3}{2}C_K n_b^{-5/2} \varepsilon + O(\varepsilon^2), \quad \delta_s(\varepsilon) - \delta_{\text{slow}} = \frac{3}{2}C_K n_s^{-5/2} \varepsilon + O(\varepsilon^2),$$

so the slow-side threshold distance is smaller by a factor $(\delta_{\text{slow}}/M)^{5/3}(1 + o(1))$.

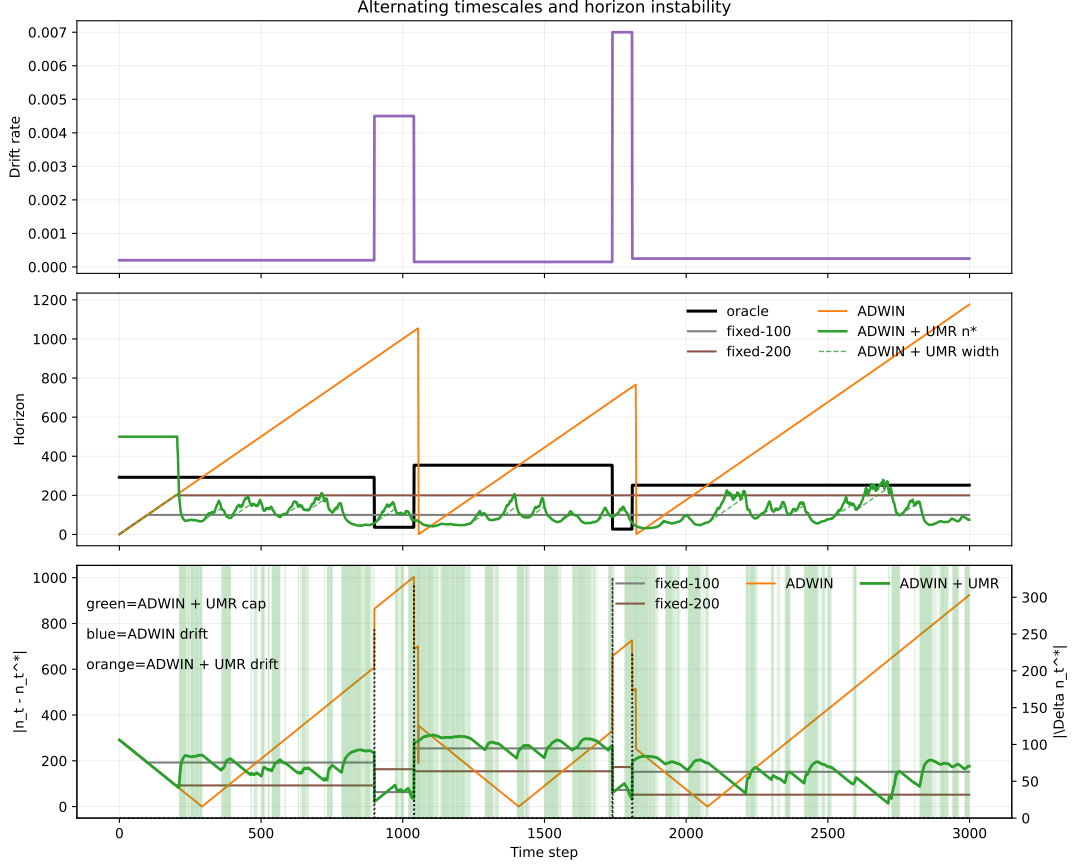


Figure 7: Alternating-timescales instability. The top panel shows the drift schedule, the middle panel compares the local oracle horizon with fixed windows and ADWIN+UMR, and the bottom panel shows horizon regret over time. The figure shows that contraction is easier to trigger than re-expansion after a low-drift stretch.

Proof. The EMA recursion has the closed form

$$\hat{\zeta}_{t_0+k} = \Delta^* + (\hat{\zeta}_{t_0} - \Delta^*)q^k$$

whenever the regime is constant with fixed point Δ^* . Hence after the transition from δ_{slow} to M ,

$$\hat{\zeta}_{t_0+k} = M - (M - \delta_{\text{slow}})q^k,$$

and after the return to the slow regime,

$$\hat{\zeta}_{t_0+T_{\text{burst}}+k} = \delta_{\text{slow}} + (\hat{\zeta}_{\text{end}} - \delta_{\text{slow}})q^k.$$

Because $n^*(\delta)$ is monotone decreasing on the unclipped interval and its inverse is $g(n) = C_K n^{-3/2}$, the condition $|n^*(\hat{\zeta}_{t_0+k}) - n_b| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+k} \geq \delta_b(\varepsilon)$, which gives the displayed formula for $T_{\text{contract}}(\varepsilon)$. Likewise, $|n^*(\hat{\zeta}_{t_0+T_{\text{burst}}+k}) - n_s| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+T_{\text{burst}}+k} \leq \delta_s(\varepsilon)$, which yields $T_{\text{expand}}(\varepsilon)$.

The inequality $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$ holds whenever the burst lasts long enough that

$$1 - q^{T_{\text{burst}}} \geq \rho_\varepsilon,$$

since this makes the numerator of T_{expand} larger than that of T_{contract} by the exact threshold relation above. The small- ε expansions follow by Taylor expansion of $n \mapsto C_K n^{-3/2}$ around n_b and n_s . $\square$

Remark 3.2 (Effect of clipping). If clipping at $n_{\min}$ is active near the burst side, then contraction can only be faster. The observed asymmetry is therefore conservative: clipping weakens the theoretical barrier in the burst regime, but it cannot eliminate the longer slow- side recovery. In streams with gradual bursts, the observed ratio can exceed the step-burst threshold because $\hat{\zeta}_{\text{end}}$ does not reach M .

3.4 Horizon Misalignment Carries Cost

The next question is when horizon misalignment becomes operationally expensive. For each backend, we compare its instantaneous horizon n_t with the local horizon target n_t^* and plot the excess error $E_t - E_t^*$ against both the absolute gap $|n_t - n_t^*|$ and the relative gap $|\log(n_t/n_t^*)|$. Here E_t^* is the best error achieved at time t by the fixed-window oracle grid, and the curve is binned over all seeds and time points.

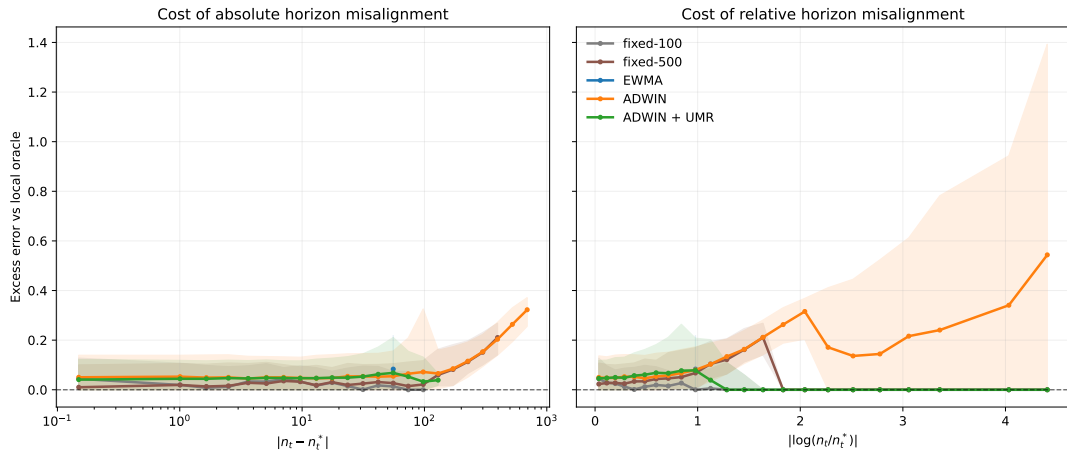


Figure 8: Excess error as a function of horizon misalignment. The relation is backend dependent rather than universal: fixed-100 stays inside a low-cost band, whereas fixed-500 and ADWIN pay substantially more once the gap widens. ADWIN+UMR sits between these extremes.

These results expose a measurable tolerance band around the useful-memory scale, together with backend-specific degradation once that band is exceeded.

3.5 Representative Real-Stream Case Study

The synthetic experiments carry the clearest identification, but ELEC2 shows the same qualitative phenomenon on public data. Coarse horizon adaptation still changes the operating regime outside the synthetic setting, although the effect is smaller and more backend dependent. A Bikes summary is given in appendix F.

Additional stability and sensitivity results appear in the appendix: bootstrap intervals and EMA-sensitivity in Appendix C, phase-wise recovery in Appendix B, the Bikes summary in Appendix F, and the controlled downstream check in Appendix G.

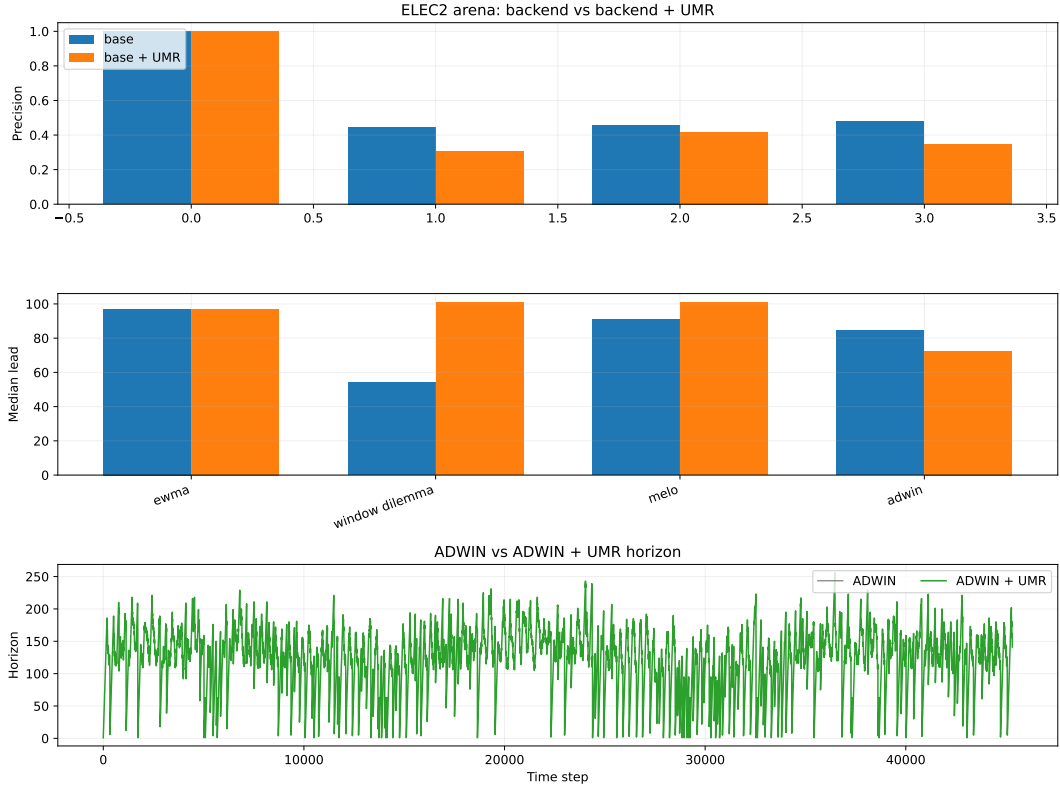


Figure 9: Public-stream case study on ELEC2. The real-stream effect is smaller and more backend dependent than in the synthetic benchmarks, but coarse horizon adaptation still changes the operating regime on public data.

3.6 What the Evidence Shows

Taken together, the evidence supports five claims. First, the finite-memory optimum is real and not a fitting artifact. Second, temporal validity and changepoint evidence are distinct objects. Third, the cap-only regime is operationally visible. Fourth, horizon recovery is asymmetric under alternating timescales. Fifth, horizon control has operational value, but its predictive payoff depends on the backend and on the drift regime.

4 Related Work

Tracking under nonstationarity. Dynamic regret, online least-squares, and adaptive filtering ask how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. Here the same question is kept in distributional form, with the memory horizon as the main object of analysis.

Variation budgets and horizon selection. The $\zeta^{1/3}$ exponent also appears in non-stationary stochastic optimization and bandits [Besbes et al., 2015, Cheung et al., 2019, Wang et al., 2024, Yuan and Lamperski, 2020]. Those results concern gradient-based methods on convex losses in Euclidean space. Here, drift is W_2 -

Lipschitz on measures, the estimator class is causal averaging kernels, and Proposition 2.9 makes the $\zeta^{1/3}$ rate a minimax lower bound for arbitrary window-restricted estimators.

The same cube-root tuning shows up in sliding-window analyses of drifting data [Bifet and Gavalda, 2007]: the window length is the control variable, but the quantity being protected is temporal validity rather than regret alone. The same perspective appears here, but with the geometry changed from scalar losses to distributional tracking.

Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze a related drift-aware weighting problem in Wasserstein distributionally robust optimization [Keehan et al., 2025]. The present focus is narrower: when memory itself is finite, how long can retained evidence remain useful before age becomes the dominant source of error?

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. In our setting, geometry quantifies how much old evidence can be retained before it becomes stale.

Adaptive forgetting and detection. EWMA, forgetting-factor RLS, and Kalman filtering all balance variance reduction against responsiveness [Jazwinski, 1970]. The cube-root law makes temporal validity explicit. The asymmetry result is adjacent to changepoint detection and EMA-based control, but the contribution here is the gap between those questions: once drift evidence is mapped into a horizon rule, re-expansion can be slower than contraction.

This also connects to the concept-drift literature more broadly [Gama et al., 2004, 2014]. In particular, the “Window Dilemma” argument of Hinder et al. [2023] emphasizes that drift detection is ill-posed when the window scale is itself uncertain. The same issue appears here in a different form: temporal validity is not fixed in advance, so the horizon becomes part of the object that must be estimated.

Public-stream baselines. Window-Dilemma-style and MELO-style baselines widen the empirical comparison. The comparison is broader, but the claim stays the same: the cube-root regulator is a worst-case horizon law for the class studied here, not a claim about every possible drift-path class.

Multi-horizon ensembles attack the same practical issue from another angle by maintaining several historical windows and combining their predictions through uncertainty or expert weighting. The public-stream experiments use that family only as a neighboring design point, not as the main object of analysis [Rajput et al., 2025].

Freshness and Age of Information. The Age-of-Information literature asks how old the last received update is and how update policies should balance communication cost against information staleness [Kaul et al., 2012, Yates et al., 2021]. The distributional analogue is the same question under W_2 drift: useful memory has

a finite worst-case horizon, and the cube-root law quantifies that worst-case scale for the class studied here.

5 Limitations and Scope

Scope of the law. The cube-root law is exact for the finite-memory averaging class studied here, and the lower bound is stated for window-restricted estimators on a W_2 -Lipschitz drift class. The lower bound itself is proved through a Gaussian location subclass and endpoint W_2 error, so its structural claim is worst-case and subclass-based rather than a universal theorem over all possible online estimators.

More refined drift classes may admit tighter staleness growth and different horizon exponents. The analysis here isolates the linear-staleness law induced by the W_2 -Lipschitz path bound.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle comes mostly from drift estimation. At the same time, a short observable calibration prefix can materially improve the regulator, so drift estimation is part of the method rather than a hidden nuisance parameter. Better local geometry estimators may therefore sharpen horizon recovery without changing the law.

Geometry beyond the current metric. The formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments show. The current experiments are not a leaderboard over drift detectors. The public stream results show the same operational distinction seen in the synthetic benchmarks: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

Scope of the comparison suite. The benchmark suite now includes a Window-Dilemma-style switcher and a MELO-style multi-horizon hedge, but it is still not a full survey of adaptive tracking methods. The analysis remains centered on the worst-case cube-root law and on the temporal-validity interpretation of memory.

The drift proxy itself has a practical resolution limit: very small block sizes can grossly overestimate $\hat{\zeta}_t$, so the calibration prefix needs blocks large enough to smooth local noise. In the current sweep, $d = 50$ is acceptable, while $d = 100$ is effectively stable.

On the current ELEC2 and Bikes arenas, adding UMR to the multi-horizon hedge changes results only modestly and in a benchmark-dependent way; it acts as a horizon prior, not a universal replacement for uncertainty weighting.

Path-class refinement. The linear staleness term comes from the local W_2 -Lipschitz path bound and is therefore a worst-case statement for that class. Characterizing narrower drift classes in which temporal misalignment accumulates more slowly, together with matching lower bounds, remains open.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? Under the drift class studied here, useful memory has a finite temporal horizon, and in the worst-case linear-staleness regime that horizon obeys a cube-root law. The governing object is a variance-staleness trade-off: preserving temporally valid evidence requires limiting the lag induced by old memory.

The theoretical contribution is a decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is a horizon regulator: UMR turns the worst-case law into a temporal-validity cap, which keeps the effective horizon from drifting far into stale-memory territory.

The strongest empirical lesson is that horizon scale, horizon tracking, and predictive payoff are distinct objects. Horizon misalignment carries a nonlinear operational cost, but the transfer from gap to error depends on the backend and on the drift regime.

Empirically, the signature regime is ‘cap-only’: the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff here. The alternating-timescales benchmark adds a secondary lesson: contraction is easier to induce than re-expansion, so EMA-driven horizon regulators have a recoverability gap.

ELEC2 is the representative real-stream case, with a short Bikes summary in the appendix. Together they show that the temporal-validity gap is not synthetic-only.

Finite-memory systems are best judged by how long their retained evidence remains current enough to matter. The design consequence is simple: regulate useful memory and track the age at which evidence stops being current.

Useful memory under drift has a finite temporal horizon, but its predictive value is policy-dependent.

The cube-root exponent is tied to the worst-case linear-staleness envelope studied here. Characterizing how this horizon changes under narrower stochastic path classes is the next step.

Code and reproducibility artifacts are available in the project repository [Parreira, 2026].

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Backend + UMR Protocol

When a backend is paired with the useful-memory regulator (UMR), the runtime protocol is a shared backend-agnostic horizon cap: it estimates a local drift scale, smooths that estimate with an EMA, and converts it into a runtime horizon

bound. The shared drift proxy and horizon rule are implemented in `drift/umr.py`. The canonical manuscript-facing entry points are the generation scripts in `code/`, which write final outputs to `artifacts/figures/`, `artifacts/csv/`, and `artifacts/tables/`.

A.1 Shared Regulator

At each time step, the regulator maintains a local drift estimate and a target horizon:

$$\Delta_t = \frac{|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}|}{d}, \quad \hat{\zeta}_t = \alpha \Delta_t + (1 - \alpha) \hat{\zeta}_{t-1},$$

$$n_t^* = \text{clip} \left(\left(\frac{C_K}{\max(\hat{\zeta}_t, 10^{-9})} \right)^{2/3}, n_{\min}, n_{\max} \right).$$

There is no explicit cooldown parameter. The only stabilization is clipping through $n_{\min}$ and $n_{\max}$, plus the EMA smoothing in $\hat{\zeta}_t$.

A.2 Horizon-Capped Backends

For estimators whose natural state is an adjustable window size or effective horizon (sliding-window mean, EWMA, Window-Dilemma-style switching, MELO-style expert mixing), the runtime horizon is simply clipped by the regulator:

$$h_t^{\text{eff}} = \min(h_t^{\text{backend}}, n_t^*).$$

The backend then computes its estimate using h_t^{eff} exactly as it would have used h_t^{backend} .

Uncapped fixed-window baselines are separate controls: they hold their window length constant throughout the run and do not receive the regulator.

A.3 Detector Wrapper Instance

One detector-side instantiation uses ADWIN on a scalar signal. The wrapper maintains a rolling buffer of the last $n_{\max}$ observations and applies the same cap logic to the detector width:

1. Update the drift proxy and compute n_t^* .
2. Feed the new observation to ADWIN.
3. Read the current ADWIN width w_t .
4. If $w_t > n_t^*$, record a cap event, reset ADWIN, and re-feed the most recent n_t^* buffered observations to the fresh detector.
5. If the detector then reports drift, record a warning.

The cap event is therefore a horizon event, not a detector event. The detector warning is still ADWIN’s own decision after the reset path. This wrapper is one instance of the shared UMR rule, not the defining form of the regulator.

A.4 Hyperparameters

Three parameter families are used.

Table 2: Synthetic benchmark defaults used by code/generate_cuberoot_adwin.py.

Scenario	Default hyperparameters
Constant	<code>drift = 0.001, fixed_window = 100, fixed_long_window = 500, ewma_alpha = 0.05, adwin_delta = 0.002, Ck = 1.0, drift_window = 100</code>
Piecewise	<code>piecewise_drifts = (0.0005, 0.003, 0.001)</code> with the same window, EWMA, ADWIN, and C_K settings as above
Alternating	<code>piecewise_drifts = (0.0002, 0.0045, 0.00015, 0.007, 0.00025), piecewise_lengths = (900, 140, 700, 70, 1190), fixed_long_window = 200</code> , and the same remaining defaults
Common	<code>$n_{\min} = 10, n_{\max} = 500$</code>

Synthetic core suite.

Real-stream arenas. The ELEC2 and Bikes arenas are generated by code/generate_elec2.py and code/generate_bikes.py with:

- `fixed_window = 100, fixed_short_window = 50, fixed_long_window = 300;`
- `ewma_alpha = 0.05;`
- `block_size = 24, ema_alpha = 0.03;`
- `min_window = 30, max_window = 300, scale = 1.25;`
- `baseline_window = 100, prefix_length = 2000;`
- `adwin_delta = 0.03;`
- Page-Hinkley event detector: $\delta = 0.01$, threshold 84.0, $\alpha = 0.9999$;
- Window-Dilemma windows (50, 100, 300) with quantiles (0.33, 0.67);
- MELO-style expert windows (30, 50, 100, 200, 300) with learning rate 6.0 and discount 0.995.

Drift-proxy calibration. When C_K is not fixed externally, the code estimates it from a stationary prefix via `calibrate_umr_constant`, i.e. standard deviation times the square root of prefix length. Appendix E additionally reports the prefix-validated (d, α) sensitivity study, generated by code/generate_zeta_support.py and code/generate_zeta_self_calibration.py; that robustness analysis does not replace the fixed-regulator core.

A.5 Algorithm

Algorithm 1. UMR update rule.

1. **Input:** stream $x_{1:T}$, backend B , prefix length m , block size d , EMA coefficient α , bounds $[n_{\min}, n_{\max}]$, constant C_K .
2. For each $t = 1, \dots, T$:
 - (a) Update Δ_t from the last two blocks of length d .
 - (b) Update $\hat{\zeta}_t = \alpha\Delta_t + (1 - \alpha)\hat{\zeta}_{t-1}$.
 - (c) Compute $n_t^* = \text{clip}((C_K / \max(\hat{\zeta}_t, 10^{-9}))^{2/3}, n_{\min}, n_{\max})$.
 - (d) If B is horizon-based, set the effective horizon to $\min(h_t^B, n_t^*)$ and evaluate the backend with that horizon.
 - (e) If B is a detector wrapper, update the detector; if its internal width exceeds n_t^* , reset it and re-feed the last n_t^* buffered observations; then record any detector warning.
3. **Output:** backend estimate, warnings, n_t^* , drift proxy, and cap events.

Figure 10: UMR update rule.

The scripts in `code/` instantiate this protocol and write the resulting artifacts to the canonical output directories under `artifacts/`.

B Phase-Wise Recovery Details

The phase-wise calibration check for the UMR-regulated ADWIN benchmark shows that the same calibration stays in the right operational range when the drift regime changes discretely.

Table 3: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Regulated n^*	Oracle MAE	ADWIN + UMR MAE
0.0005	200	191.2	0.0741	0.0874
0.0030	50	58.1	0.1258	0.1381
0.0010	100	89.7	0.0876	0.1033

C Supplementary Empirical Checks

Bootstrap intervals, transition-window summaries, and EMA sensitivity checks support the same qualitative conclusions as the primary empirical results.

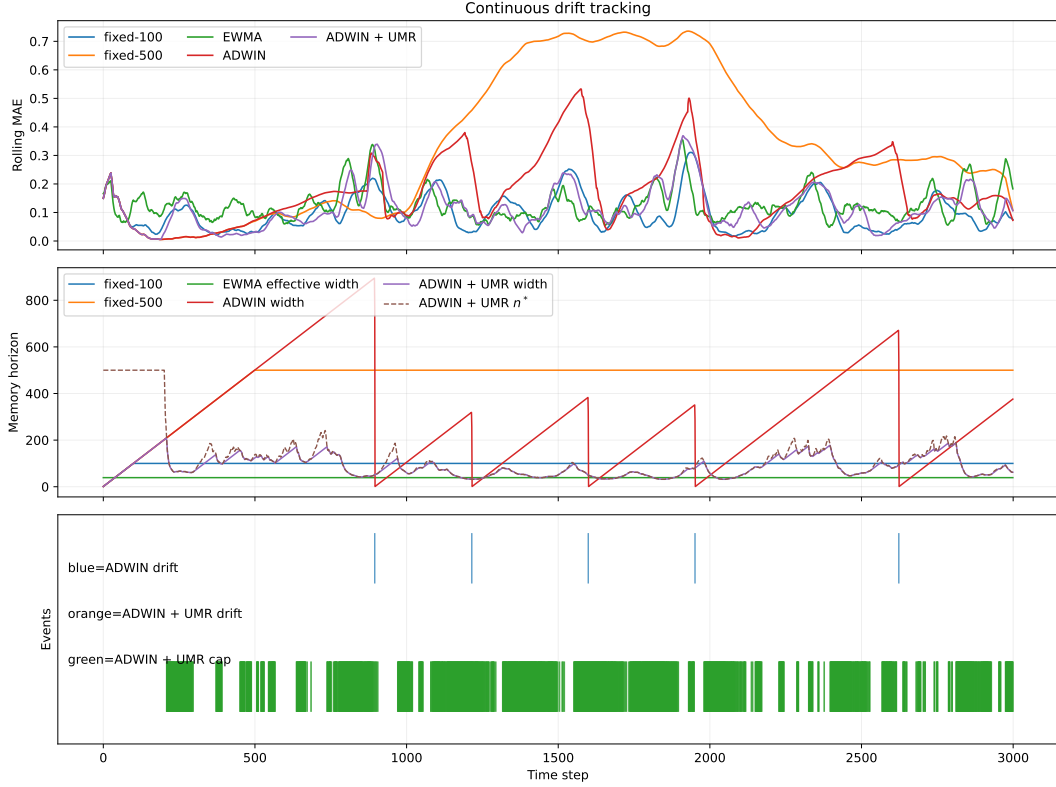


Figure 11: Piecewise-drift recovery. The same regulator calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime.

C.1 Bootstrap Stability Check

Table 4: Seed-wise bootstrap summary for the main synthetic benchmark. The first column block reports tail-MAE means with 95% bootstrap intervals across the 20 seeds; the second block reports paired differences versus ADWIN+ UMR.

Method	Tail MAE [95% CI]	Δ vs ADWIN+ UMR [95% CI]
Fixed-100	0.0849 [0.0815, 0.0884]	-0.0129 [-0.0143, -0.0115]
Fixed-500	0.2388 [0.2275, 0.2500]	+0.1409 [+0.1319, +0.1502]
EWMA	0.1291 [0.1263, 0.1319]	+0.0313 [+0.0278, +0.0350]
ADWIN	0.1757 [0.1658, 0.1852]	+0.0779 [+0.0706, +0.0857]
ADWIN + UMR	0.0978 [0.0943, 0.1016]	—

Table 5: Mean horizon regret over the first 50 steps after each regime change. Contraction averages over the two fast-drift transitions, and expansion averages over the two slow-drift recoveries. Lower is better.

Method	Contraction	Expansion
fixed-100	68.0	203.1
fixed-200	168.0	103.1
ADWIN	789.5	477.7
ADWIN + UMR	67.1	248.1

C.2 Transition-Window Asymmetry Table

C.3 Drift-EMA Sensitivity

The recovery asymmetry persists when the drift estimator itself is varied. The next sweep changes the EMA coefficient α in the drift estimator while keeping the alternating-timescales benchmark fixed. Larger α makes the drift estimate more reactive, reducing contraction regret while expansion regret stays persistently larger; the expansion-to-contraction ratio remains above one throughout the sweep and is higher at the larger α values.

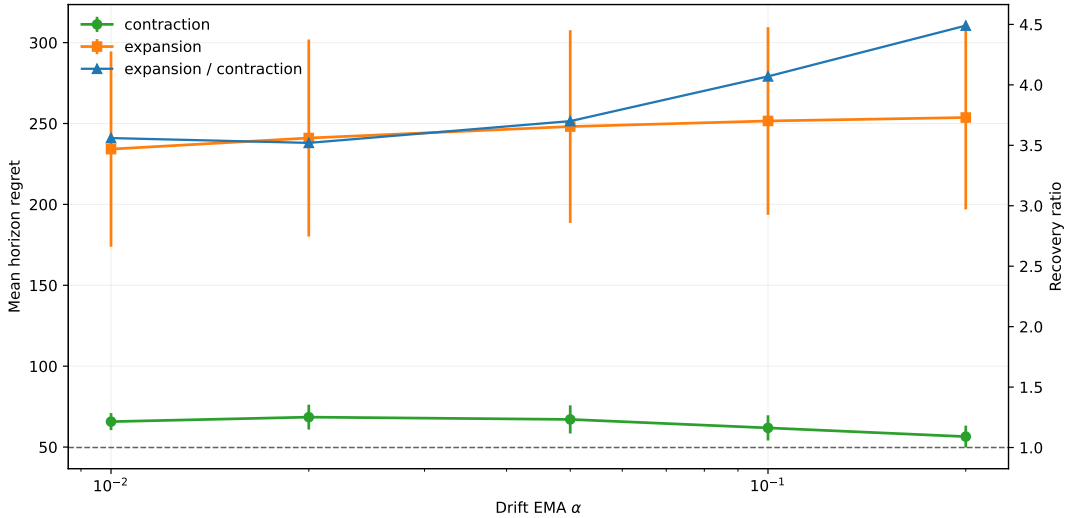


Figure 12: EMA sensitivity of horizon recovery on alternating timescales. The drift-EMA coefficient changes how quickly the regulator reacts to bursts, but the asymmetry remains: expansion regret stays above contraction regret across the sweep.

D Finite-Sample Geometry Check

For the finite-sample term in lemma 2.2, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate

dimension, then fit a log-log slope in n . We also check the Sinkhorn proxy in one dimension at a few fixed values of ε .

Table 6: Empirical scaling check for the finite-sample term. The bounded-support W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, and the slope degrades as intrinsic dimension increases. The Sinkhorn proxy is reported at fixed ε , where the measured slope and constant both vary with regularization.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

The numerical check is only meant to support the modeling choice behind lemma 2.2. In low dimension, the raw Wasserstein term is compatible with an $n^{-1/2}$ fluctuation scale, whereas higher-dimensional raw Wasserstein decays more slowly. For fixed ε , the Sinkhorn proxy acts as a measurement layer whose calibration constant changes with regularization and ambient geometry even when the horizon argument keeps the same formal shape.

E Online Drift Estimation and Sensitivity

The dynamic horizon regulator uses a short-window drift proxy built from block differences. For block length d , let $\bar{x}_t^{(d)}$ denote the mean of the most recent block and $\bar{x}_{t-d}^{(d)}$ the mean of the previous block. The instantaneous drift proxy is

$$\Delta_t = \frac{|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}|}{d},$$

and the smoothed estimate follows the EMA recursion

$$\hat{\zeta}_t = \alpha \Delta_t + (1 - \alpha) \hat{\zeta}_{t-1}.$$

The regulator then sets

$$n_t^* = \text{clip} \left(\left(\frac{C_K}{\max(\hat{\zeta}_t, 10^{-9})} \right)^{2/3}, n_{\min}, n_{\max} \right).$$

Table 7 reports the sensitivity sweep on the alternating-timescales benchmark. Larger α makes the horizon react faster to bursts, which lowers contraction regret but leaves expansion regret larger; the expansion-to-contraction ratio stays above one throughout.

The concrete estimator used in the benchmarks and the accompanying empirical sweep show that the asymmetry persists as the drift filter is retuned.

Table 7: Drift-EMA sensitivity on the alternating-timescales benchmark. The ratio remains above one across the sweep, so re-expansion is slower than contraction for every tested α .

α	Contraction	Expansion	Ratio	Tail MAE
0.01	65.73	234.18	3.56	0.0938
0.02	68.51	241.03	3.52	0.0923
0.05	67.10	248.12	3.70	0.0925
0.1	61.86	251.57	4.07	0.0934
0.2	56.48	253.70	4.49	0.0940

E.1 Calibration Cost of C_K Misestimation

Let the regulator use an estimated constant $\hat{C}_K = rC_K$ while the true tracking law is governed by C_K . Since the selected horizon is $n^*(\hat{C}_K) = (\hat{C}_K/\zeta)^{2/3}$, the excess tracking error relative to the oracle choice is

Proposition E.1 (Calibration Cost of C_K Misestimation).

$$\Delta E(r) = \mathcal{E}(n^*(\hat{C}_K); C_K, \zeta) - \mathcal{E}(n^*(C_K); C_K, \zeta) = C_K^{2/3} \zeta^{1/3} \left(r^{-1/3} + \frac{1}{2} r^{2/3} - \frac{3}{2} \right).$$

Moreover, for $r = 1 + \delta$ with $|\delta| \ll 1$,

$$\Delta E(r) = \frac{1}{9} E_{\min} \delta^2 + O(\delta^3),$$

so small calibration errors are locally quadratic and symmetric to second order.

Table 8: Local calibration-cost sensitivity for the horizon constant. The excess error is normalized by $E_{\min}$, so the table is scale-free.

$r = \hat{C}_K/C_K$	Exact $\Delta E/E_{\min}$	Comment
0.90	0.0018	modest underestimation already moves the horizon
0.95	0.0004	local error is very small near the optimum
1.00	0.0000	oracle calibration
1.05	0.0004	local error is very small near the optimum
1.10	0.0015	moderate overestimation remains mild
1.25	0.0085	larger bias becomes visible
1.50	0.0288	large bias moves the operating point

The practical implication is simple: moderate calibration error in C_K is benign to first order, but large systematic bias still moves the operating point materially. The calibration-prefix machinery in Appendix A and the block-size sensitivity table below quantify how this shows up in practice. A short numerical check of the proposition is generated by `code/generate_zeta_support.py`.

E.2 Calibration Sensitivity Summary

Two short sensitivity checks are enough to capture the practical behavior of the regulator. First, the Sinkhorn regularization level changes the constant but not the exponent: across the tested values, the observed $\zeta^{1/3}$ slope stays at approximately 0.333, while the implied C_K moves from about σ at large ε to roughly 3.3σ at small ε . Second, the block length must not be too small: $d = 10$ produces a very large positive bias in the drift proxy, $d = 50$ is already usable, and $d = 100$ is effectively stable.

The practical behavior is governed by two sensitivities. First, the Sinkhorn regularization level changes the calibration constant C_K without altering the cube-root scaling argument. Second, the block length must not be too small: overly short blocks make the drift proxy noisy and bias $\hat{\zeta}_t$ upward, which in turn pushes the regulator toward under-memory.

E.3 Robustness of the EMA Drift Proxy

Let the block-difference proxy satisfy $\Delta_t = \zeta_t + \xi_t$, where ξ_t is a zero-mean noise term with $\text{Var}(\xi_t) \leq \sigma_\Delta^2$ and the EMA recursion is

$$\hat{\zeta}_t = \alpha \Delta_t + (1 - \alpha) \hat{\zeta}_{t-1}.$$

If $\zeta_t \equiv \zeta$ is constant, then

$$\mathbb{E}[\hat{\zeta}_t] = \zeta + (1 - \alpha)^t (\hat{\zeta}_0 - \zeta), \quad \text{Var}(\hat{\zeta}_t) \leq \frac{\alpha}{2 - \alpha} \sigma_\Delta^2.$$

So smaller α reduces variance but increases lag, while larger α reacts faster but amplifies noise.

Since the regulator uses

$$n_t^* = \left(\frac{C_K}{\hat{\zeta}_t} \right)^{2/3},$$

the relative horizon error is first-order sensitive to relative drift-estimation error:

$$\frac{\delta n_t^*}{n^*} \approx -\frac{2}{3} \frac{\delta \hat{\zeta}_t}{\zeta}.$$

Thus persistent overestimation of ζ induces under-memory, and persistent underestimation induces over-memory. If the proxy has multiplicative bias $\mathbb{E}[\Delta_t] = (1 + b)\zeta$ with $b > 0$, then asymptotically

$$\frac{\hat{n}^*}{n^*} = (1 + b)^{-2/3},$$

so a 10% overestimate of drift shrinks the target horizon by about 6.5%.

For persistent overestimation, the same calculation gives systematic under-memory. For example, with a 10% positive bias the asymptotic horizon ratio is $1.1^{-2/3} \approx 0.94$, and with a 25% bias it is about 0.86.

Table 9: EMA robustness check for a constant drift proxy with additive noise. The empirical mean matches the target drift, the empirical variance matches the closed-form $\alpha/(2 - \alpha)$ law, and the implied horizon distortion stays close to one when the proxy is unbiased.

α	$\mathbb{E}[\hat{\zeta}_t]$	$\text{Var}(\hat{\zeta}_t)$	Theory	$\mathbb{E}[\hat{n}^*]/n^*$
0.01	1.0002	0.00032	0.00031	1.0000
0.02	0.9999	0.00062	0.00063	1.0004
0.05	1.0004	0.00161	0.00160	1.0006
0.1	1.0001	0.00330	0.00329	1.0018
0.2	1.0000	0.00693	0.00694	1.0039

E.4 Prefix-Validated Auto-Calibration

The operational fix is to choose (d, α) on a short calibration prefix by minimizing an *observable* prequential score over a small grid. For each candidate pair (d, α) , run the regulated estimator on the prefix, generate the one-step-ahead forecast $\hat{x}_{t|t-1}^{(d, \alpha)}$, and score it by

$$S(d, \alpha) = \frac{1}{T_0 - 1} \sum_{t=1}^{T_0-1} (x_{t+1} - \hat{x}_{t|t-1}^{(d, \alpha)})^2.$$

The selected pair minimizes this prequential MSE on the prefix and is then used for the full run. This is implementable online because it only uses observed data and the regulator’s own forecasts.

Table 10: Prefix-validated auto-calibration on the synthetic benchmarks. The calibration prefix chooses (d, α) by observable one-step-ahead prequential error, then the selected parameters are run on the full stream.

Scenario	Mean α	Mean d	Tail MAE (default)	Tail MAE (calibrated)
Constant drift	0.07	83.8	0.0978	0.1015
Alternating drift	0.03	93.8	0.0925	0.0952

The failure mode is also visible: if the proxy is overreactive, the selected window contracts too aggressively and under-memory appears in the tail; if it is too inertial, the horizon lags the drift and over-memory follows. The calibration rule makes that sensitivity explicit by choosing the grid point that best predicts the next-step observations on the prefix.

The implemented self-calibrating rule is therefore simple: keep a small grid of block scales and EMA coefficients, score each candidate by prequential one-step MSE on a short prefix, then deploy the best candidate for the remainder of the stream. For non-Lipschitz local behavior, the grid should include at least one short and one longer block scale so that bursty changes do not force a single timescale to carry the entire burden. This is an operational refinement, not a replacement for the fixed-regulator results above.

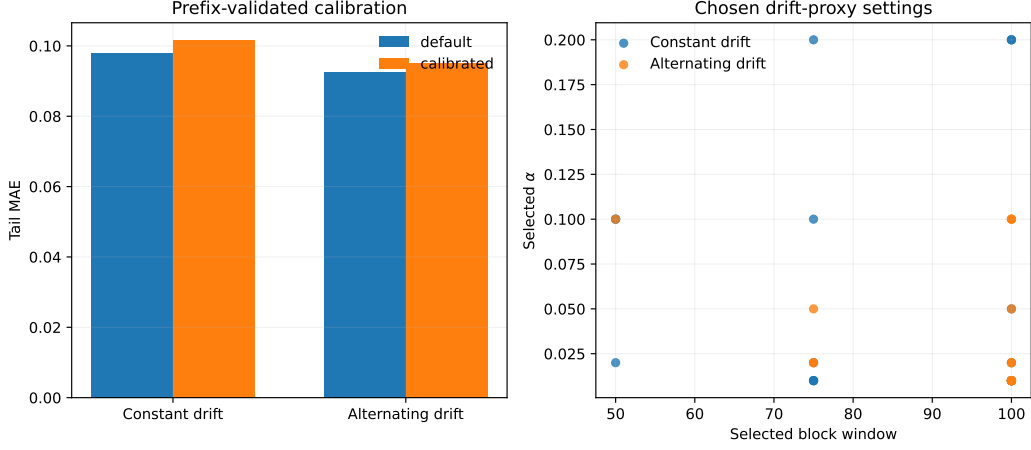


Figure 13: Prefix-validated self-calibration of the drift proxy. The selected (d, α) shifts the operating point of the regulator; on these scenarios it does not improve tail MAE relative to the fixed default.

F Additional Public-Stream Summary

Bikes provides a second public-stream summary and shows that the same temporal-validity gap is not unique to one stream.

Table 11: Bikes public-stream summary. Precision is matched-warning precision against Page-Hinkley events; median lead is the median warning lead in steps.

Method	Precision	Median lead	Leads
EWMA	0.146	132.0	59
EWMA + UMR	0.146	132.0	59
Window Dilemma	0.127	108.5	76
Window Dilemma + UMR	0.128	125.0	76
MELO-style	0.141	147.5	76
MELO-style + UMR	0.129	147.0	71
ADWIN	0.526	92.0	251
ADWIN + UMR	0.520	82.0	141

The qualitative picture is the same as in ELEC2: coarse horizon adaptation still matters on public data, but the payoff remains backend-dependent.

G Controlled Downstream Check

The controlled synthetic downstream benchmark focuses on the phase-2 cap-only window: UMR has already contracted, while ADWIN remains silent.

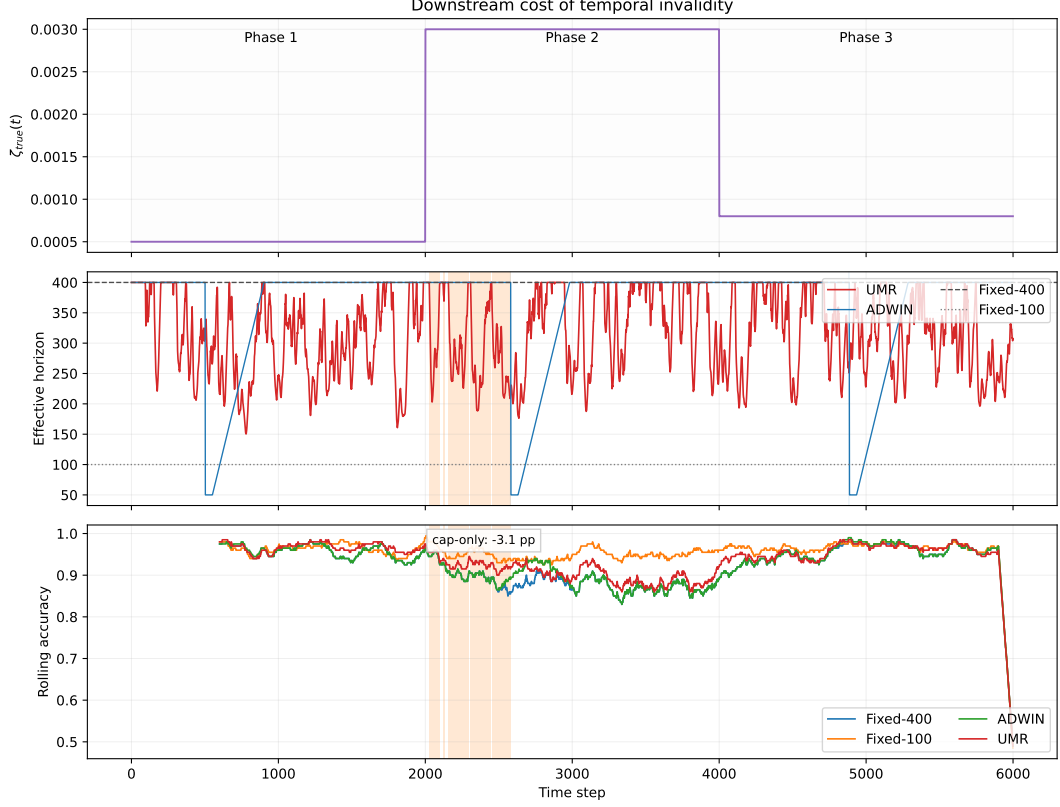


Figure 14: Controlled downstream benchmark. Top: true drift rate. Middle: effective horizons and the cap-only interval. Bottom: rolling prequential accuracy. The representative seed shows a 3.12 point gain for UMR over Fixed-400 within the cap-only window.

Table 12: Controlled downstream benchmark summary. The reported delta is UMR minus Fixed-400 within the cap-only interval, averaged across six seeds.

Method	Global acc.	Cap-only acc.	Δ cap-only	Mean cap-only width
Fixed-400	0.9304	0.9040	+0.00	293.3
Fixed-100	0.9598	0.9576	+5.36	293.3
ADWIN	0.9340	0.9040	+0.00	293.3
UMR	0.9415	0.9205	+1.66	293.3

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